

RESEARCH ARTICLE

Assessments of population exposure to environmental pollutants using air quality measurements during Commonwealth Games-2010

Dilip Chate, G. Beig, Trupti Satpute, Saroj Kumar Sahu, K. Ali, Neha Parkhi, and Sachin Ghude

Indian Institute of Tropical Meteorology, Pune, India

Abstract

During the “Commonwealth Games” 2010 (CWG-2010) in Delhi, the Indian government has implemented an ambitious project “System of Air quality Forecasting And Research (SAFAR)” for monitoring and forecasting air-quality scenario. Using high-precision spatio-temporal measurements of criteria pollutants from the SAFAR network, the number of cases are estimated for total, cardiovascular and respiratory mortalities and hospital admissions. In a thinly populated airport area, the excess number of cases for total mortality show ~10 for PM_{2.5} and 25 for PM₁₀, whereas, ~110 for PM_{2.5} and ~300 for PM₁₀ in most populous Delhi University (DU) area. Cardiovascular mortality in airport area show ~5 and <10 for PM_{2.5} and PM₁₀, respectively, but, in DU area show ~55 for PM_{2.5} and ~140 for PM₁₀. In DU locality, respiratory mortality shows ~7 and ~20 for PM_{2.5} and PM₁₀ and, hospital admissions show ~11 and ~30 for PM_{2.5} and PM₁₀, respectively. In airport area, excess cases of respiratory mortality and hospital admission tends to one for exposure to PM_{2.5} or PM₁₀ levels indicating effective exposure is the key factor for health hazards. As public health gains, low air pollution levels were observed before the CWG due to effective washout by monsoonal rain and during CWG under policy-induced air quality measures could increase the life expectancy as against to post-CWG period. These results are important for the megacities in developing world as the SAFAR project is internationally recognized by the Global Urban Research Meteorology and Environment of the World Meteorological Organization.

Keywords

Air-quality, CWG-2010, excess number of cases, megacity, relative risk, SAFAR

History

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Introduction

In a typical urban environment, the population is often exposed to a wide variety of air pollutants on time scales of days to years. The linkages between respiratory health impacts and air-quality deterioration are documented in the literature for both the developed and developing world (Chate, 2011; Chhabra et al., 2001; Cohen et al., 2005; Gurjar et al., 2010; Guttikunda & Goel, 2013; Pope et al., 2002; Pope & Dockery, 2006). For annual time scales, Gurjar et al. (2010) reported very low excess cases of total mortality from air-quality data of suspended particulate matter (SPM measurements refers to particles diameter up to 100 µm), SO₂ and NO₂ in some megacities (Los Angeles, New York, Osaka Kobe, Sao Paulo and Tokyo), while, a large number of excess cases were found for Dhaka, Beijing, Karachi, Cairo and Delhi. Chhabra et al. (2001) examined the respiratory health of 4000 adults living in the areas across the nine air pollution monitoring sites in Delhi. Recently, Foster & Kumar (2011) quantified health effects of air-quality regulations in Delhi using air pollution data of 113 sites (spread across Delhi and its neighboring areas) from July to December 2003 for

exposure of 3989 persons to air pollution. Data of air pollution used in most of these studies were after the conversion of commercial vehicles to compressed natural gas (CNG) in Delhi, and the closure of polluting industries in residential areas from 2000 to 2002. They suggest that the changes in air-quality with regulations in Delhi had a substantial effect on respiratory health, and the effect was of concern among low-expenditure households. Beside, the results based on aggregated air pollution data, the aforementioned studies were limited in the quality of measurements due to unavailability of adequate sets of the real time aerosols and gas analyzers and hence could not represent a realistic air-quality scenario in Delhi.

Many other epidemiological studies (Valavanidis et al., 2008 and references therein) in the past three decades found a better exposure-response relationship between particulate matter for short-term effects (premature mortality, hospital admissions) and cumulative health effects (morbidity, lung cancer, cardiovascular and cardiopulmonary diseases). The evidence for population exposures to PM_{2.5} and PM₁₀ indicates that there are risks to health at concentrations currently found in many cities in developed and developing countries (WHO, 2006). The short-term health effects of particulate and gaseous air pollutants have been well documented for the developed world, mainly through time-series studies relating short-term elevations in ambient pollutants

Address for correspondence: Dilip Chate, System of Air Quality And Forecasting Research, Dr. Homi Bhabha Road, NCL Post Office, Pune 411008, India. E-mail: chate@tropmet.res.in

levels to increases in morbidity and mortality. Results of 124 studies of the largest cities in North America and Europe showed an increase in the rate of death from any cause ranging from 0.2% to 0.6% for an increase in ambient PM_{10} concentrations of $10 \mu\text{g m}^{-3}$ (Samoli et al., 2008). The health risks associated with short-term exposures to PM_{10} and $\text{PM}_{2.5}$ are likely to be similar in megacities in developed and developing countries. It is advantageous to have estimations for short-term population exposure based on precise air-quality measurements from dense network of monitoring stations covering large area such as during “Commonwealth Games 2010 (CWG-2010)” in Delhi for a representative area specific population and air pollutant levels.

Globally, Delhi is the seventh most populous megacity in the world, which has been declared by the National Cancer Registry Program as the most carcinogenic megacity in India (ICMR, 2005; UN HABITAT, 2008). The population of Delhi including National Capital Region of Delhi (Delhi-NCR) is ~ 16.7 million, exhibiting a decadal growth of 20.96% with 11 297 persons per km^2 (Census of India, 2011). In Delhi, to date, the CWG-2010 (3–14 October) was the largest multi-sport event, where a large number of international tourists, sport persons and spectators joined the sport events for which relatively clean air was maintained with much reduced emissions from industry, road traffic and construction activities. Separate lanes were dedicated for the athletes to cut down on vehicular emissions along those corridors. Norm of 50 ppm sulfur in petrol/diesel was implemented (from 1 April 2010), new CNG buses/radio taxis were fleetted, heavy vehicles were banned from entering Delhi and additional metro-trains were arranged for transportations. Coal-based power plants were closed down and rooftop solar power plant setup at Thyagaraja Stadium provided green power during CWG-2010 period and also 15 Government buildings were retrofitted. Energy Conservation Building Code was adopted for all new buildings having connected load of 500 KW or more. About five million potted plants were planted for CWG 2010. Also, 9 new city forests were created and 300 parks were rejuvenated which increased green area from 289 to 500 km^2 in Delhi and Delhi-NCR. With the timely implementation of aforementioned policy-induced measures for CWG-2010, the Delhi's air quality could be improved (Ali et al., 2013) which provided a rare opportunity for health effects study using network observations. Under a mission mode project “System of Air quality Forecasting And Research” (SAFAR), the dense network of Air Quality Monitoring Stations (AQMS) and Automatic Weather Station (AWS) was setup for CWG-2010. With the coordinated efforts of the Indian Government and researchers, an improved air-quality was maintained and monitored throughout the CWG-period. By adopting a methodology described in the World Health Organization (WHO) reports (WHO, 1999, 2000, 2003, 2006) and air quality data obtained from a network of AQMS stations (SAFAR), total, cardiovascular and respiratory mortality and hospital admissions are estimated for short-term (CWG-period) effective population exposure to $\text{PM}_{2.5}$, PM_{10} , NO_2 and O_3 levels. The population exposure is considered for spatially distributed air pollutants within areas of 2 km radii from each of the AQMS stations located

in residential, airport and sport complexes for the estimations of total, cardiovascular and respiratory mortality and hospital admissions. Spatial distribution of pollutants in a typical source-receptor system (Chate, 2011) determines the effective population exposure within areas of 2 km radii from each of the AQMS stations. This exposure is crucial factor for the overall health effects which mainly depends on number density of persons or the effective population exposure to air pollutant levels in the source-receptor system. Health outcomes increase with the number of persons under exposure and time-scales of the exposure. The effective population exposure to relatively clean air over CWG-2010 period can be less adverse than that on normal days. Improvements in an air quality would results in substantial gains in public health in terms of life expectancy. The results from the developed countries show that a reduction of $10 \mu\text{g m}^{-3}$ in the ambient air concentration of particulate matter increases an average life expectancy of 0.61 years (Krewski, 2009). In this context, the health effects under the aforementioned policy-induced air-quality measures which were imposed exclusively for CWG-2010 would set an example for gains in public health as against the results available in the literature for Delhi.

Study area

Observations of various air pollutants were made at 11 AQMS established in and around Delhi under SAFAR project in the area of 2500 km^2 including 900 km^2 in Delhi-NCR. More details on SAFAR air-quality monitoring setup can be found at <http://safar.tropmet.res.in/>. Total seven AQMS-AWS stations located in different directions (representative of Delhi and downtown Delhi-NCR) were operated round the clock during CWG-2010 period. As seen in Figure 1, Indian Institute of Tropical Meteorology, Delhi (IITMD) (1) is a residential area, away from the major traffic insertions. Major Dhyani Chand National Stadium (MDNS) (4) located at the India Gate is close to the major roads. Thyagaraj (6) and CWG Village (CWGV) (7) are sport complexes. Delhi University (DU) (8) is $\sim 500 \text{ m}$ away from the main traffic roads and IGI (9) is an airport and Talkatora Garden (10) is a sport complex, hereafter they will be referred as (1), (4), (6), (7), (8), (9) and (10).

Measurements of the criteria pollutants (PM_{10} , $\text{PM}_{2.5}$, O_3 and NO_2) were made at $\sim 3 \text{ m}$ above ground, at each of the AQMS stations (1, 4, 6, 7, 8, 9 and 10) consisting of online analysers during CWG-2010 period. The analysers were operated continuously and the data were recorded at every 5 min interval. The raw data were processed for quality check, so after filtering outliers, $>90\%$ validated data were subsequently averaged to 1-h interval. Ozone was measured with a photometric UV analyser and NO_2 with the chemiluminescence's NO_x analyser. PM_{10} and $\text{PM}_{2.5}$ were monitored by Beta Met One BAM1020 (BETA attenuation) aerosol analysers. Calibration of ozone analysers were done on every alternate day using inbuilt O_3 calibrators, whereas calibrations of NO_x analysers were performed using external calibration cylinders with multipoint calibration technique. Also, meteorological parameters (wind speed, wind direction, temperature and relative humidity), total solar radiation were recorded by AWS systems at 5 minute intervals.

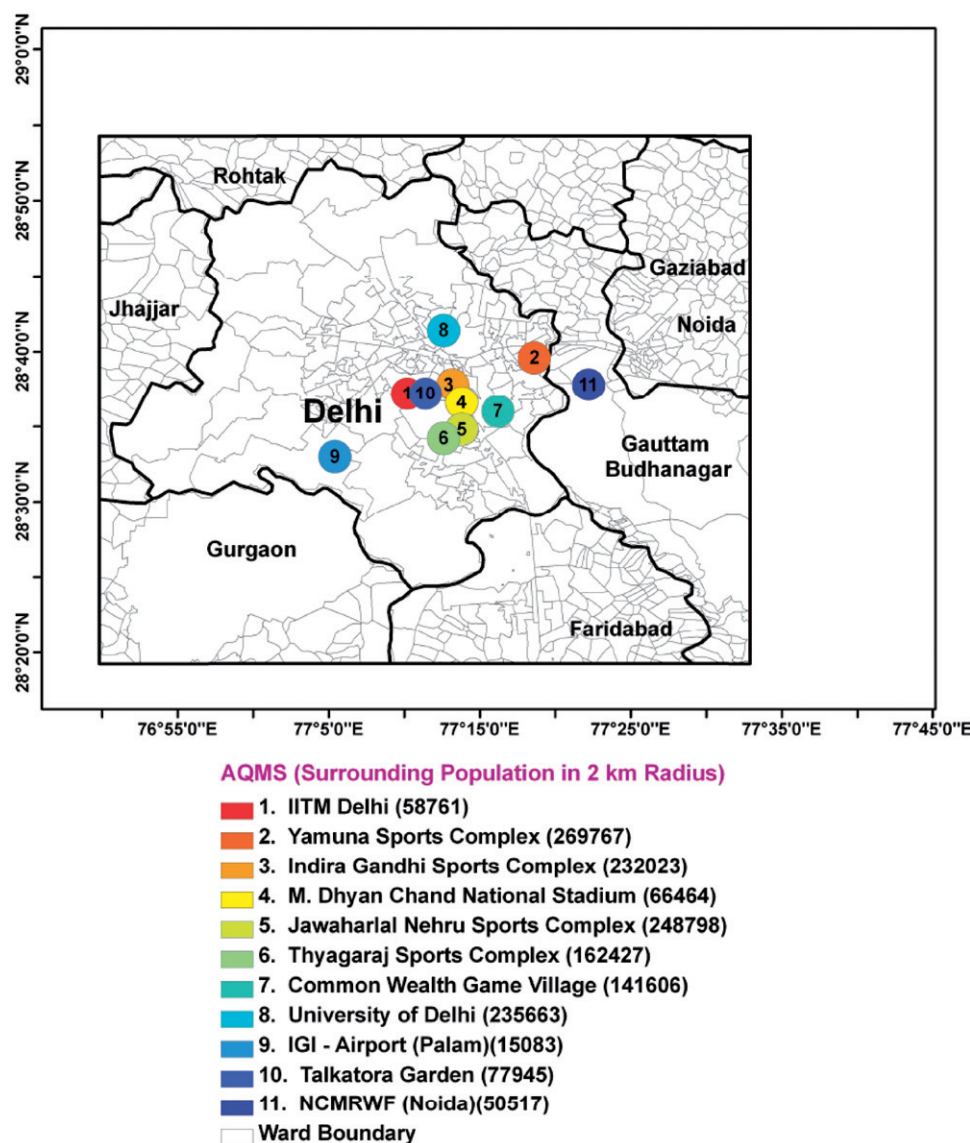


Figure 1. Population distributions within 2 km radius around the AQMS for the year 2010.

Air quality guidelines for $PM_{2.5}$, PM_{10} , O_3 and NO_2

The time response of the health effects of the pollutants has been taken into account in defining the National Ambient Air Quality Standards (NAAQS) and WHO Air Quality Guidelines (AQG). The NAAQS values for continuously monitored $PM_{2.5}$ and PM_{10} are set to 60 and $100 \mu g m^{-3}$, respectively, over 24-h averages for the Indian conditions (http://cpcb.nic.in/National_Ambient_Air_Quality_Standards.php). In case of $PM_{2.5}$ and PM_{10} particles, evidence shows that, even at very low concentrations, health effects can be observed for effective population exposures, thus they have no safe level at which no risk to health can be assumed. We adopt 60 and $100 \mu g m^{-3}$ as threshold limits for $PM_{2.5}$ and PM_{10} , respectively, and concentration–response relationships and methodology for a population attributable-risk proportion (AP) from the WHO reports (2000, 2003, 2006). With evidence from both chamber and field studies indicates that there is considerable individual variation in response to O_3 , provides a good case for reducing the WHO's AQG for O_3 from the existing level $120\text{--}100 \mu g m^{-3}$. As per the existing

WHO's AQG, short-term NO_2 value assumed to be $200 \mu g m^{-3}$ (WHO, 2000, 2003, 2006). For O_3 and NO_2 threshold levels, the WHO AQG limits are assumed to be background values. If the average O_3 level $<100 \mu g m^{-3}$ and NO_2 level $<200 \mu g m^{-3}$, no adverse health effects occur for their short-term exposure to healthy persons. However, long-term population exposure to pollutants increases the risk of no accidental mortality independent of age, gender and geographic region. NO_2 being a poorly water-soluble gas, deposit far more peripherally in a respiratory tract, but does not reach the alveoli in any significant amounts. Also, the daily maxima for O_3 and NO_2 over CWG-2010 period (3–14 October 2010) in all the AQMS areas showed levels below the limits of WHO recommended AQG for both O_3 and NO_2 (WHO, 2000, 2003, 2006). Therefore, the health effects due to short-term population exposure to O_3 and NO_2 concentrations can be neglected at least for the CWG-2010 period. Although some symptoms and harmful effects can be expected for the vulnerable people, children and asthmatic people, however, such specific health impact assessments are beyond a scope of the present study. We present the results for total,

cardiovascular and respiratory mortalities and hospital admissions by analyzing the PM_{2.5} and PM₁₀ data of the 7 AQMS stations (1, 4, 6, 7, 8, 9 and 10) obtained during CWG-2010 period and population distributions over study areas within 2 km radii from each of the AQMS stations (Figure 1).

Methodology

The concentrations of PM₁₀ and PM_{2.5} equivalent to NAAQ recommendations are assumed to be their background levels for which the relative risk RR(c) is set at 1. Assumed WHO air quality default values of RR(c) changes by 3%, 2%, 8% and 4.4% for each 10 µg m⁻³ linear increase in either PM₁₀ or PM_{2.5} levels for total, cardiovascular, and respiratory mortalities and hospital admissions, respectively (WHO, 1999, 2000, 2003, 2006). Further, corresponding baseline incidences (per 100 000) are assumed to be 1013, 497, 66 and 101.4 for these four respective health effects (Gurjar et al., 2010 and references therein). Quantification of the health impacts caused by exposure to a particular air pollutant is based on the population AP concept (Douwes et al., 2002; Rothman et al., 2008). The AP is the fraction of health impacts (WHO, 1999, 2000, 2003, 2006) which can be attributed to the exposure in a given population can be calculated using Equation (1),

$$AP = \frac{\sum[\{RR(c) - 1\} \times p(c)]}{\sum[\{RR(c) \times p(c)\}]} \quad (1)$$

where RR(c) = relative risk for the selected health outcome, which can be derived from the exposure-response as a function of rate of increase of PM_{2.5} and PM₁₀ levels, and p(c) = proportion of the population within 2 km radii of each of the AQMS locations. We used daily data (the arithmetic means of PM_{2.5} and PM₁₀ concentrations) of seven monitoring stations for RR(c) estimations. Knowing the RR(c) at the observed PM_{2.5} and PM₁₀ levels (summarized in Table 1) and the corresponding p(c) values, AP (% of total, cardiovascular and respiratory mortalities and hospital admissions) can be estimated using Equation (1).

Assuming a certain baseline frequency (NAAQS values of daily averages of PM_{2.5} and PM₁₀) of selected health outcome in the population (I) of size N within 2 km radii of each of the AQMS-AWS systems, the rate (or number of cases per unit

population of each of the seven AQMS locations) attributed to the exposure in population (IE) to PM_{2.5} and PM₁₀ levels can be calculated (WHO, 1999, 2000, 2003, 2006) with Equation (2) as below.

$$IE = I \times AP \quad (2)$$

Consequently with Equation (2), the frequency of occurrence in the population free from exposure (INE) can be estimated using Equation (3) below,

$$INE = I - IE = I \times (1 - AP) \quad (3)$$

The estimated incidences in the non-exposed population (INE), the excess incidences [$\Delta I(c)$] and excess number of cases [$\Delta N(c)$], respectively, at PM_{2.5} and PM₁₀ levels (Table 1) of exposure (depending on locations of each of the AQMS-AWS systems in Delhi and Delhi-NCR) can be calculated using the following Equations (4) and (5), respectively, as below,

$$\Delta I(c) = \{RR(c) - 1\} \times p(c) \times INE \quad (4)$$

$$\Delta N(c) = \Delta I(c) \times N \quad (5)$$

Results and discussion

Before the commencement of the CWG-2010, Delhi was in the under heavy monsoonal rainfall. In September, rainfall was recorded in excess of the average rainfall by a 90%. Cloudy sky condition persisted for a long-duration till a week before the games when monsoon withdrawal started in Delhi. The region was completely out of monsoonal/rainfall activity during the games period. During CWG-2010 period, temperature was in the range of 30–34 °C during peak sunshine hours of the day. Night-time temperature varied between 23 °C and 26 °C. Relative humidity (RH) during 1100–1900 h (IST) was in the range of 30–50%, and during night times it is in the range of 70–95%. This scenario of RH remained nearly constant over the CWG-2010 period of observations. Major period of solar radiation reaching the Earth's surface showed variation from 220 to 400 Wm⁻² between 1000 and 1600 h.

The methodology described above depends mainly upon steady state population data of year 2010 (within 2 km radii

Table 1. Daily mean PM_{2.5} and PM₁₀ levels at various AQMS stations during CWG-2010.

CWG 2010	IITMD (1)		Major D (4)		Thyagaraj (6)		CWGV (7)		DU (8)		IGI (9)		NDMC (10)	
	PM ₁₀	PM _{2.5}	PM ₁₀	PM _{2.5}	PM ₁₀	PM _{2.5}	PM ₁₀	PM _{2.5}	PM ₁₀	PM _{2.5}	PM ₁₀	PM _{2.5}	PM ₁₀	PM _{2.5}
3 Oct	236	141	210	69	243	127	312	145	238	125	252	182	205	109
4 Oct	240	133	236	96	257	107	329	141	274	127	237	161	212	102
5 Oct	232	136	274	114	252	121	310	144	315	156	237	186	219	110
6 Oct	214	129	221	80	249	118	248	126	267	132	209	163	201	104
7 Oct	315	205	294	139	310	175	345	186	300	167	325	224	259	152
8 Oct	270	162	270	130	265	120	317	160	321	158	260	175	229	130
9 Oct	244	139	241	104	263	133	281	135	273	128	273	172	219	113
10 Oct	222	98	251	52	249	85	302	104	277	100	196	104	204	78
11 Oct	153	58	178	40	200	56	206	62	221	70	174	93	169	53
12 Oct	177	69	192	47	197	64	212	73	224	76	169	94	181	62
13 Oct	184	80	185	48	204	68	205	76	214	74	180	101	196	70
14 Oct	190	92	203	58	207	79	223	96	226	90	200	110	184	76

from each of the AQMS system) as shown in Figure 1 and ambient concentrations of $PM_{2.5}$ and PM_{10} during 12 days of sport events (3–14 October 2010). Following the $PM_{2.5}$ and PM_{10} data (Table 1), the health risks are quantified for population exposure in each of the 7AQMS-AWS locations (Figure 1). Among the AQMS-AWS localities considered, DU (8) has the largest population (~235 663) and being airport, IGI (9) is thinly populated (~15 083) area. Accordingly, number of persons available in the study areas for exposure or effective population exposures to daily average concentrations of $PM_{2.5}$ and PM_{10} in different localities (Table 1) are used to calculate the excess number of mortality and morbidity cases. The results for total, cardiovascular and respiratory mortalities and hospital admissions are shown in Figure 2(a–d) for $PM_{2.5}$ and in Figure 3(a–d) for PM_{10} , respectively. Figures

also show the error bars for the ranges in total, cardiovascular and respiratory mortalities and hospital admissions for $PM_{2.5}$ and PM_{10} , respectively. Both the figures shows increase in the number of mortalities and morbidities in the locality (4) to (8) while dip in airport area (9), and moderate number of these cases in localities (1) and (10). In general, total, cardiovascular and respiratory mortalities and hospital admissions mainly depends on effective population exposures to $PM_{2.5}$ and PM_{10} levels in each of the AQMS source-receptor system. The higher number of cases in the DU (8) area may be attributed to the large population exposure to the $PM_{2.5}$ and PM_{10} levels. The lower number of cases show for airport (9) site owing to nearby thinly populated area due to restrictions on private constructions of high-rise buildings and banned on heavy vehicles.

Figure 2. (a–d) Excess number of cases for $PM_{2.5}$ (1) IITMD (4), (6), (7), (8), (9) and (10).

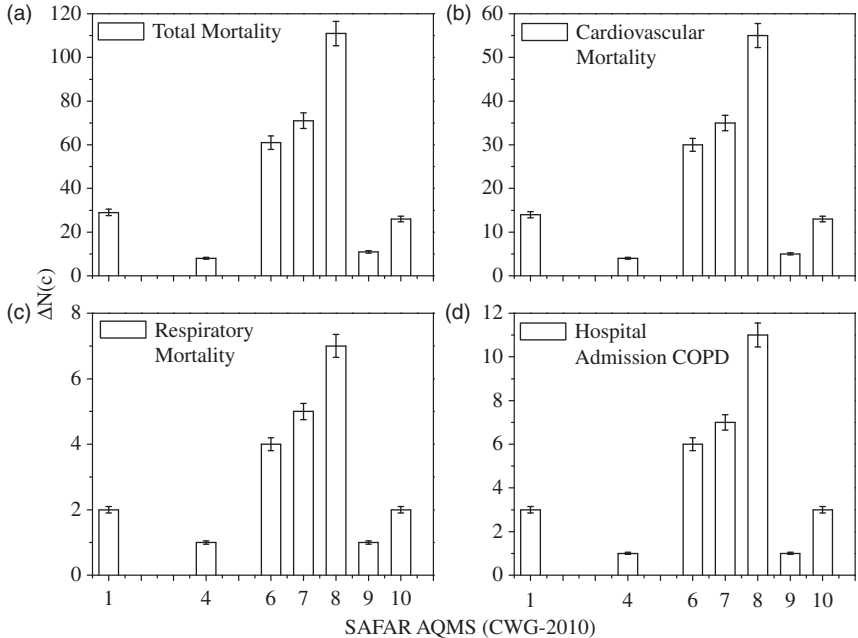
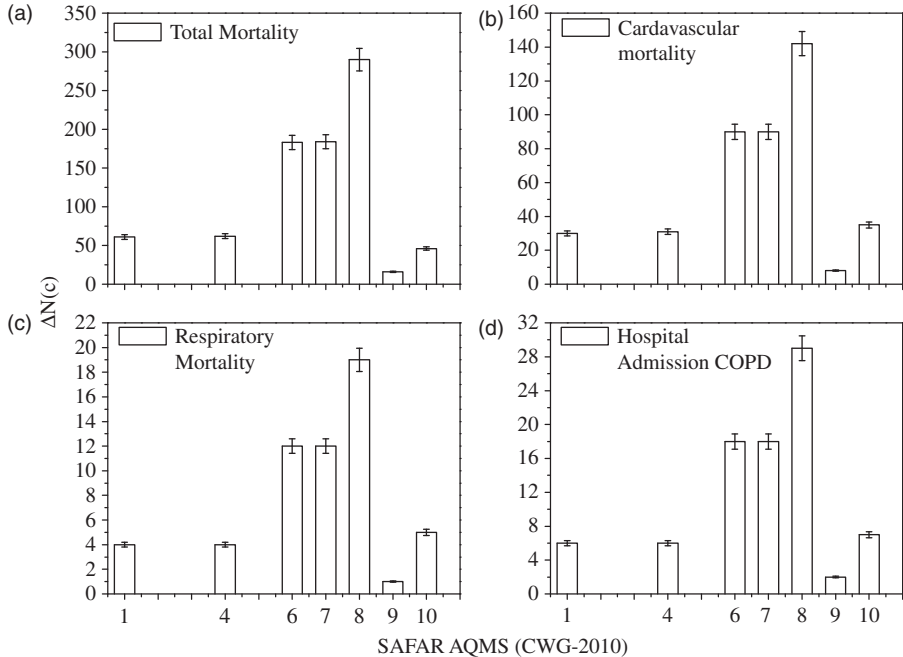


Figure 3. (a–d) Excess number of cases for PM_{10} (1) IITMD (4), (6), (7), (8), (9) and (10).



Total mortality

The excess number of no accidental deaths (i.e. total mortality) shown in Figures 2(a) and 3(a) have been calculated taking into account the $PM_{2.5}$ and PM_{10} concentration-based relative risks and the effective population exposures to the air quality in the monitoring areas. Accordingly, airport site (9) show very low number of excess cases, about 10 associated with the $PM_{2.5}$ and about 25 to PM_{10} concentrations. On the other hand, the excess number of cases in DU (8) area seen in Figures 2(a) and 3(a) are about 110 and 300 due to availability of large population for exposure to $PM_{2.5}$ and PM_{10} concentrations, respectively. Also, the excess number of cases of total mortalities in areas (6) and (7) show relatively high values than those in localities (1), (4), (9) and (10). The monitored areas (6), (7) and (8) show greater health risks due to moderate number of persons available for exposure to $PM_{2.5}$ and PM_{10} levels relative to population in the other areas. Although aforementioned health effects are of population exposure to $PM_{2.5}$ and PM_{10} levels for control environment over CWG-2010 period, they can be viewed against the health impacts associated with the aggregated data of air pollutants reported elsewhere (Gurjar et al., 2010; Guttikunda & Goel, 2013). For instance, during 1998–2005, the per million average excess number of cases for total mortality are between 750 and 930 in industrial and 720–920 for residential areas (Gurjar et al., 2010). Also, the excess number of deaths (i.e. total mortality) for entire Delhi are $\sim 10\,000$ per year. Total premature mortality due to the effects of total particulate matter concentrations for the base year 2010 in Delhi are reported in the range 7350–16200 per year (Guttikunda & Goel, 2013). It is worth to note about our results that effective population exposure to the $PM_{2.5}$ or PM_{10} concentrations is the key factor in potential health impact and do not harm persons if they are rarely get outdoor exposure to the pollutants. However, even low $PM_{2.5}$ or PM_{10} levels become relevant when large number of persons undergo exposure to these particles all the time. Thus, the most important factor in assessing the health effects is the effective population exposure to the pollutants monitored within study areas.

Cardiovascular mortality

Because of the tremendous number of people affected, the impact of air pollution on cardiovascular disease represents a serious public health problem. Figures 2(b) and 3(b) show the number of cases estimated for cardiovascular mortality owing to the exposure effects of the $PM_{2.5}$ and PM_{10} concentrations. The airport area (9) show number of cardiovascular cases ~ 5 for population exposure to the $PM_{2.5}$ and these cases show <10 for PM_{10} concentrations. In DU area (8), these cases with the population exposure to the $PM_{2.5}$ and PM_{10} concentrations show ~ 55 and ~ 140 , respectively. On the other hand, Gurjar et al. (2010) reported that per million average excess number of cases related to cardiovascular mortality ranges between 270–330 and 240–310 in industrial and residential areas, respectively, and for Delhi, it is ~ 3500 per year. The cardiovascular mortality cases are important for the fraction of total population (in Delhi) staying and having health problem related to heart and they are sensitive to air-quality of this area even in air quality maintained environment.

Cardiovascular disease constitutes a global problem and is the leading cause of death in the world which is also a major cause of disability and of reduced quality of life (WHO, 2007). People with pre-existing cardiovascular disease, diabetic and elderly individuals are also considered to be more susceptible to air pollution-mediated cardiovascular effects (Simkhovich et al., 2008).

Respiratory mortality

Figures 2(c) and 3(c) show the excess number of mortality cases estimated for respiratory problems with the population exposure to $PM_{2.5}$ and PM_{10} levels. The numbers in these cases in DU area (8) show ~ 7 and ~ 20 in consideration of effective population exposure to the $PM_{2.5}$ and PM_{10} concentrations, respectively. The airport area (9) shows one case of respiratory mortality for population exposure to $PM_{2.5}$ or PM_{10} . On the other hand, the excess number of cases per million of population are reported in the range 120–150 and 110–140 in industrial and residential areas, respectively, and also ~ 1600 per year for Delhi (Gurjar et al., 2010). The respiratory system is the gateway for 150001 of air that enter through the nostrils, and after its appropriate conditioning, arrives to the delicate net structured by the lung parenchyma. Although our results about respiratory mortality in an airport area show one and much less (7–20) in DU (8) area, but, the number of persons suffering from respiratory health problem can opt for medical advisories for their comfortable outdoor activities.

Hospital admissions

Air pollution has been implicated in Chronic Obstructive Pulmonary Disease (COPD) pathogenesis, and there is enough evidence to support the association between air pollution and COPD exacerbations and worsening of those with pre-existing COPD. $PM_{2.5}$ and PM_{10} levels are associated with its development (Salvi & Barnes, 2009). Figures 2(d) and 3(d) show the excess number of COPD cases (hospital admissions) for population exposure to the $PM_{2.5}$ and PM_{10} levels. The MDNS (4) and airport areas (9) show only 1 case of COPD for $PM_{2.5}$ and ~ 2 for PM_{10} in airport area (9). The COPD in DU area (8) show 11 and 30 for $PM_{2.5}$ and PM_{10} , respectively. The dip in COPD cases in MDNS (4) and airport (9) areas can be attributed to less number of persons available for exposure to $PM_{2.5}$ or PM_{10} as compared to those in other areas. Whereas, COPD cases based on aggregated data (Gurjar et al., 2010) shows 110–140 in industrial areas and 100–130 in residential areas and ~ 1500 per year for entire Delhi.

Considering the policy-induced control measures which improved air-quality over CWG-2010 period, the presented estimates may be considered to be lower limits. Observed levels of total SPM are measure of air-quality standards (WHO, 2000, 2003, 2006). The assumptions for concentrations of PM_{10} and $PM_{2.5}$ below their NAAQ values for 24-h averages equivalent to their background levels with relative risk ~ 1 and also, the relative risk changes for each $10\ \mu g\ m^{-3}$ increase in PM_{10} and $PM_{2.5}$ concentrations shows overall linkages of air-quality with human health risks. However, smaller particles ($<PM_{2.5}$) have high number concentrations

in observed levels of total particulate matter, large surface to volume ratios, longer atmospheric residence times, high proportion of persistent organic compounds and ability to penetrate deeper into the respiratory system (Chate, 2011 and references therein). Therefore, they can have a significant impact on mortality and morbidity, caused by respiratory and cardiovascular diseases (Chen et al., 2005; Dominici et al., 2005; Pope et al., 2002; Schwartz et al., 1996, 2001). Also, future anticipated rise in these health hazardous particles, via infrastructural growth, increased transportation, fossil fuel burning, atmospheric new particle formation, urbanizations, population growth and migrations across megacities all over the world is expected to increase health risks related to respiratory problems (Chate, 2011; Dockery & Stone, 2007; Pope et al., 2004).

Limitations

The results based on assumed WHO air quality default values of RR(c) which changes for each $10 \mu\text{g m}^{-3}$ linear increase of both PM_{10} and $\text{PM}_{2.5}$ levels for total, cardiovascular, and respiratory mortalities and hospital admissions, respectively, are presented with error bars. The relative risks for $\text{PM}_{2.5}$ and PM_{10} particles should be developed by correlating their levels and weather parameters with the health data in time-series over short-term scales. Limited by health data availability we did not determine relative risks and instead used the WHO default values. The development of the relative risks for fine ($\text{PM}_{2.5}$) and coarse ($\text{PM}_{10}-\text{PM}_{2.5}$) particles independently would provide the precise health effects as they differ in deposition functions into respiratory system when inhaled or exposed to population. For health effects results, assumption of steady state population within 2 km radii of each of the AQMS source receptor system available for exposure to $\text{PM}_{2.5}$ and PM_{10} remain valid over a CWG-2010 period. The adverse health effects are dependent on effective population exposure to the concentrations of fine and coarse particles as well as length of exposure (Pope, 2007). The health effect assessments are performed for improved air quality during CWG-2010 period. It is interesting to know how much reduction in concentrations of $\text{PM}_{2.5}$ and PM_{10} took place due to imposition of air quality measures for CWG-2010. Because a reduction of $10 \mu\text{g m}^{-3}$ in the ambient air concentration of particulate matter was associated with an estimated increase in average life expectancy of 0.61 years (Pope et al., 2009). Although analysis of life expectancy is beyond a scope this work, average $\text{PM}_{2.5}$ level over post-CWG period (15–30 October 2010) in DU (8) area was higher than that of CWG-2010 period by $40 \mu\text{g m}^{-3}$ and PM_{10} level by $50 \mu\text{g m}^{-3}$. Similarly, in airport area, average $\text{PM}_{2.5}$ level increased by $56 \mu\text{g m}^{-3}$ and average PM_{10} by $120 \mu\text{g m}^{-3}$ over post-CWG period. Further, total, cardiovascular and respiratory mortalities and hospital admissions are found to be almost twice for Deepavali-2010 in all seven AQMS systems' sites for exposure to very high $\text{PM}_{2.5}$ and PM_{10} levels due to fireworks displays as against to those of CWG-2010 period (Beig et al., 2013). The $\text{PM}_{2.5}$ and PM_{10} levels before (20 August to 28 September 2010), during and after CWG-2010 for IITMD (1) and MDNS (4) are shown in Figure 4(a) and (b) for comparison. Before the CWG, heavy monsoonal rainfall

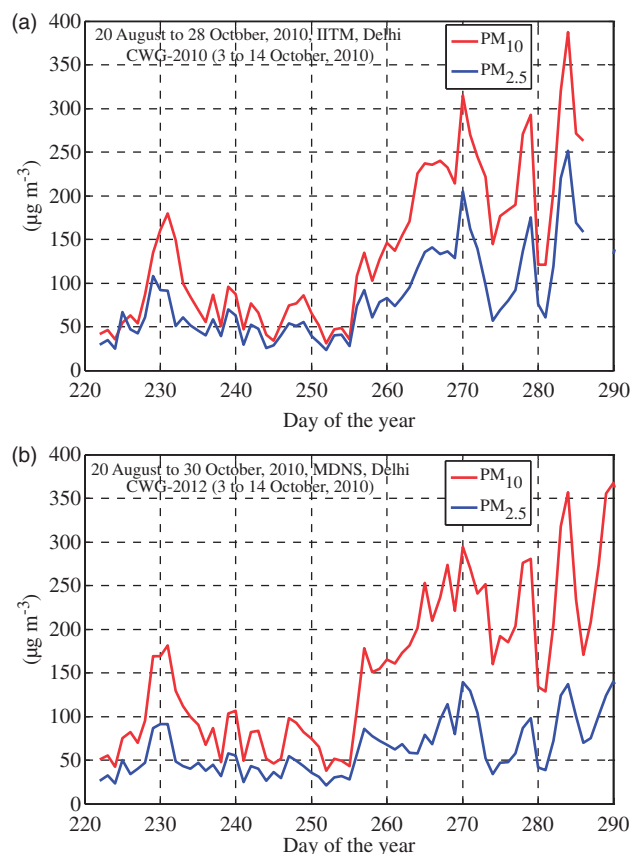


Figure 4. (a) $\text{PM}_{2.5}$ and PM_{10} levels before, during and after CWG-2010 in IITMD (1) and (b) same as (a) in MDNS (4) areas.

washed out aerosols (Tiwari et al., 2012), thus $\text{PM}_{2.5}$ and PM_{10} levels were much below than those during and after CWG. Clean air before and during CWG period could increase the life expectancy as against to post-CWG period and also to Deepavali-2010 period under very high emission due to fireworks displays (Beig et al., 2013). Long-term air pollution data from the SAFAR network would provide a systematic life expectancy analysis in line with the studies conducted for developed world.

Conclusion

This study documents the short-term health impacts of continuously monitored $\text{PM}_{2.5}$ and PM_{10} particles for effective population exposures to air quality which was maintained during CWG-2010 period under the policy-induced measures. The excess number of cases for total mortality (no accidental deaths) show an order of magnitude less in an airport area than that in DU area for the effective population exposure to $\text{PM}_{2.5}$ or PM_{10} levels. Similar patterns show for cardiovascular and respiratory mortalities as well as for hospital admissions with much less number of cases in these areas. The results indicates that exposure is the key factor in potential health impact in terms of estimations of excess number of cases for total, cardiovascular and respiratory mortalities and hospital admissions. It means high $\text{PM}_{2.5}$ or PM_{10} concentrations do not harm persons if they are rarely or very few available for exposure to these particles in the areas under question for health risks. Being the airport, this thinly

populated area show one to two excess number of cases for both the respiratory mortality and hospital admissions for population exposure to PM_{2.5} or PM₁₀ levels. On the contrary, even low PM_{2.5} or PM₁₀ levels become relevant when exposures take place effectively to large population to these particles all the time.

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Declaration of interest

The authors report no conflicts of interest. The authors alone are responsible for the content and writing of this article.

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